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Exploring Large Language Models for Scientific Question Answering via Natural Language to SPARQL Translation

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Translating scientific questions expressed in natural language into SPARQL queries that can be executed over knowledge graphs remains a significant challenge in the field of question answering. Recently, several prominent benchmarks, notably SciQA and DBLP-QuAD, have emerged to evaluate performance in this domain. In this paper, we provide a comprehensive analysis of the performance of language models on these benchmarks, assessing various optimization strategies. Our results indicate that the combined use of fine-tuning and prompting techniques, especially when incorporating strategic few-shot selection, produces excellent results on both benchmarks. These findings underscore an urgent need for more challenging benchmarks to better assess model capabilities. We identify key insights, common error patterns, and potential opportunities for transfer learning, and we discuss their implications for optimizing the performance of large language models in knowledge graph-based question answering tasks.

CCS Concepts: • **Computing methodologies** → **Machine translation**; **Information extraction**; **Natural language generation**.

Additional Key Words and Phrases: Knowledge graphs, Question answering, Language models, Fine-tuning, Few-shot learning

1 INTRODUCTION

Knowledge graphs have witnessed increased popularity in recent decades, serving not only as repositories for factual information but also as retrieval backends for large language models (LLMs) [28, 53]. Most knowledge graphs currently adopt relatively simple semantic structures, and the question-answering (QA) benchmarks are mainly designed for such structures and rely on encyclopedic knowledge graphs such as DBpedia [43] and Wikidata [69].

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Recently, a few novel benchmarks, such as *SciQA* [4] and *DBLP-QuAD* [6], have emerged, focusing instead on granular and specialized representations of scientific knowledge. This is a significant shift from the focus on general knowledge to a more detailed representation of science, posing a new set of challenges for future QA technologies. Specifically, these benchmarks address the task of translating scientific questions expressed in natural language into SPARQL queries that can be executed over domain-specific knowledge graphs. The SciQA ORKG benchmark [4] targets the Open Research Knowledge Graph[1] (ORKG)[62], which provides a detailed description of scholarly artefacts. Auer et al. [4] highlight its challenging nature by demonstrating that both knowledge graph question answering (KGQA) systems and ChatGPT (with the GPT-3.5 backend) achieve relatively low performance on this benchmark. The DBLP-QuAD benchmark [6] focuses instead on the DBLP Knowledge Graph[2], a machine-readable version of the well-known DBLP Computer Science Bibliography[3]. Unlike SciQA, it also provides, as input, the entities and relationships referenced in the queries.

The primary objective of this paper is to provide a comprehensive analysis of the performance of LLMs in the automatic translation of scientific questions into SPARQL queries, using the SciQA and DBLP-QuAD benchmarks. Specifically, we aim to assess the accuracy of various LLMs in translating natural language questions into SPARQL queries, employing a range of optimization techniques.

The most common approaches for enhancing the performance of a language model on a given task involve appropriate prompting and fine-tuning. In this study, we explore the impact of both. If a language model can effectively be prompted to solve the task, it suggests that the model inherently possesses the capability to address the problem, but may require additional context to optimize its next-token predictions [19, 74]. Consequently, we first examined the influence of zero- and few-shot learning techniques on SciQA performance. For few-shot learning, we also investigate several example selection strategies, leveraging criteria such as semantic similarity [47], diversity [39, 44], and entropy [30]. Our results show that different configurations of LLMs, utilizing both fine-tuning and few-shot learning approaches, achieve notable success on the SciQA benchmark. In particular, the fine-tuned version of Mistral-7B-v0.1 obtained the highest F1 scores while the fine-tuned version of Llama3.1-8B achieved the highest number of exact matches. Notably, some of the smaller models (T5, Dolly-v2-3b, GPT-2-large) also exhibited remarkable performance when fine-tuned. This suggests that fine-tuned low-resource models can not only match but potentially surpass the performance of larger, more resource-intensive models even in complex tasks such as scientific question answering. The error analysis revealed that T5, Llama3.1-8B, and GPT-3.5 struggled most with understanding the underlying ontological schema, whereas Mistral-7B-v0.1 exhibited a higher incidence of semantic misunderstandings, which are errors arising from an incorrect interpretation of the input questions.

To gain a deeper understanding of the capabilities of LLMs in this domain, we further evaluated the top-performing models using the *DBLP-QuAD* benchmark, which incorporates relevant entities and relations. The results indicate that Mistral-7B-v0.1, Llama3.1-8B, and T5 achieve strong performance when supplied with this type of structured information. Additionally, we evaluated these models using execution-based metrics, providing further insights into their practical effectiveness. Subsequently, we explored a retrieval-augmented ensemble approach and found that it consistently outperforms each individual model on the *DBLP-QuAD* dataset. Finally, we investigated the potential for transfer learning between SciQA and DBLP-QuAD. Specifically, we evaluated whether training on one benchmark improves performance on the other, and whether training on a combination of both datasets leads to improved results on their respective test sets. Results indicate some feasibility, but further research is needed to refine dataset integration and optimize knowledge transfer.

In summary, the paper makes the following contributions:

[1]Open Research Knowledge Graph - https://orkg.org
[2]DBLP Knowledge Graph -https://blog.dblp.org/tag/knowledge-graph/
[3]https://dblp.org/

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1) We present a performance analysis of six language models on the SciQA benchmark using three strategies: zero-shot learning, few-shot learning, and fine-tuning. In addition, we investigate eight alternative approaches to sample selection for few-shot learning.
2) We demonstrate that optimizing LLMs through effective prompting and fine-tuning techniques can achieve exceptional results on SciQA, including an F1-score of over 97.7% and exact query match accuracy exceeding 95.9%;
3) We report an in-depth error analysis of the top-performing methods, identifying distinct error patterns that need to be addressed by the scientific community in this domain.
4) We present additional experiments on the DBLP-QuAD benchmark, which show similarly strong results, yielding an F1-score of over 98.8%. These experiments also demonstrated that explicitly providing the entities and relationships referenced in natural language questions significantly benefits models, such as T5, which may otherwise struggle to learn the underlying ontological schema.
5) We report an execution-based evaluation on DBLP-QuAD, which offers additional insights into the practical applicability of the models.
6) We introduce a retrieval-augmented ensemble method and demonstrate that it outperforms all individual models on the DBLP-QuAD dataset.
7) We discuss a series of transfer learning experiments, evaluating various combinations of training and test sets across SciQA and DBLP-QuAD.
8) We release the full codebase for the experiments, along with a new version of SciQA that incorporates input entities and relations, facilitating further analysis of LLM performance in this space.

The rest of this paper is organized as follows. Section 2 explores related works, covering knowledge graph question answering and optimizations for LLM few-shot techniques. In Section 3, we provide details about the SciQA benchmark and the LLM models used in our analysis. Section 4 outlines the experimental setup and the metrics utilized for evaluation. Section 5 presents the analysis of SciQA, along with a detailed error analysis. Section 6 describes the additional experiments performed on DBLP-QuAD. Section 7 discusses the limitations of our study. Finally, Section 8 concludes the paper and outlines potential future directions.

2 RELATED WORK

In this section, we review the current state of the art regarding the semantic parsing for knowledge graph question-answering (Section 2.1) and LLM few-shot optimization (Section 2.2).

2.1 Semantic Parsing for Question Answering on Knowledge Graphs

Semantic parsing is the task of converting a natural language utterance into a machine-understandable representation, or logical form [35]. Semantic parsing can be employed for various applications, such as machine translation, code generation [73], automated reasoning, and question-answering. The resulting representation can be considered successfully converted when it accurately captures the original input's intent in natural language. The literature presents a variety of approaches for knowledge graph question answering (KGQA) [5, 17, 58, 59]. Traditional approaches usually consist of a combination of template extraction, feature engineering, Combinatory Categorial Grammars (CCG), rule-based systems, and conventional semantic parsing methods [8, 57, 66]. Neural-network-based approaches have also demonstrated their effectiveness for the KGQA task recently [17]. These approaches include simple neural embedding models [10], attention based recurrent models [21], and memory-augmented neural controller architectures [45, 51, 72]. For instance, Chakraborty et al. [17] present a work that uses neural network methods for semantic parsing of knowledge graphs. Instead, Rongali et al. [58] propose an encoder-decoder model (also referred to as sequence-to-sequence models), in which the encoder

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converts the natural language utterance into an intermediate representation, and then the decoder uses such a representation to produce its logical form.

A recent architecture is the pointer-generator networks [5] which integrates the features of both encoder-decoder models and pointer networks [68]. These networks can copy specific input segments, such as data values, to the output. These methods achieve high accuracy when trained with large and high-quality dataset, but the difficulties in acquiring such data motivate the interest in adopting strategies that can be trained with less data. Rony et al. [59] also investigate unsupervised approaches, which are able to achieve promising results. Nevertheless, these methods have scalability issues for real-world scenarios and present challenges in terms of easy adaptability or tunability for practical applications.

In contrast, LLM families such as GPT [14, 52], Llama [64], Mistral [31], and Dolly [22] demonstrate strong potential for semantic parsing with little to no training examples, particularly as advancements continue to reduce inference costs. Nevertheless, LLMs do not inherently support KG-specific parsing without access to entities in the underlying knowledge graph. Therefore, they need augmentation with a method for retrieving entities. For instance, Lehmann et al. [42] leveraged LLMs for KGQA semantic parsing, observing that training data requirements are considerably reduced when employing languages with restricted grammar and vocabulary (also known as controlled natural languages) as targets for KGQA. In parallel, there has been a substantial proliferation of KGs in scientific and academic domains, enabling advanced question-answering and a range of domain-specific tasks such as literature analysis [16, 71], automated literature review generation [9], academic writing and citation enhancement [13, 15], supporting analytical dashboards [3, 70], automated research hypothesis generation [11, 12, 36], the development of knowledge organization systems [60], and the creation of specialized conversational agents [41, 49], among others. These include KGs that describe the metadata of research publications (e.g., SemOpenAlex [27], AIDA-KG [2], DBLP KG [1], OpenCitations [23]), as well as KGs that focus on the content of these publications, capturing key entities and concepts (e.g., ORKG [62], AI-KG [26], CS-KG [25], Nano-publications [37], SoftwareKG [61]). We also observe several efforts aimed at building [24, 34] and refining [50] knowledge graphs in this domain, increasingly leveraging LLMs supported by a human-in-the-loop component [65].

The team behind the SciQA benchmark also organized the Scholarly QALD Challenge[4] as a satellite event of the 22nd International Semantic Web Conference (ISWC 2023). This led to the development of some well-performing dedicated approaches for SciQA [33, 54, 63], which typically incorporate additional components for integrating a representation of the ORKG ontological schema or identifying domain-specific entities. In contrast, our study aims to examine the efficacy of conventional LLMs when enhanced through fine-tuning and prompting techniques.

2.2 LLM Few-Shot Optimization

A few-shot optimization for LLM is a technique that enhances the performance of an LLM on a specific task by giving it a few examples of the desired input and output, and then modifying the model's parameters or the examples themselves to optimize the task accuracy. This challenges the conventional paradigm of machine learning training algorithms that require large amounts of sample data.

The main idea is to train these models to perform new tasks with just a few examples, and in the literature, we can see different methods for optimizing the quality of the examples given to the model. Kumagai et al. [40] proposed a few-shot learning method that utilizes a feature selector and a decoder. The selector employs Concrete Random Variables to select relevant features for the decoder to restore the original features of unseen instances.

Bansal et al. [7] proposed a sampling method for optimization-based meta-learning across diverse tasks with varying numbers of classes. Levi et al. [44] proposed a method for sample selection in in-context learning, showing

[4]Scholarly QALD Challenge - https://kgqa.github.io/scholarly-QALD-challenge/2023/; QALD stands for "Question Answering over Linked Data".

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that diverse samples both with and without fine-tuning yield improved performance. Likewise, Lin et al. [46] have explored few- and zero-shot learning capabilities of LLMs across a broad spectrum of tasks (e.g., paraphrasing, commonsense reasoning) with a diverse selection of examples reaching higher values of performance. Finally, Iter et al. [30] used the Cross-Entropy Difference method through a fine-tuning approach, achieving promising results and showing that it fosters an efficient usage of the parameters, thus enabling the training of smaller models on training data.

Within this paper, we present a comprehensive application of the discussed methods to the task of formalizing natural language queries. Our specific focus is on translating natural language questions into SPARQL queries, within the academic knowledge domain.

3 BACKGROUND

This section provides the necessary background for the study, detailing the SciQA benchmark and introducing the language models used in our experiments.

3.1 The SciQA dataset

The SciQA dataset[5] includes 2,565 pairs, each consisting of a natural language question and its corresponding SPARQL query. The SPARQL queries were crafted to answer the question by retrieving relevant information from the Open Research Knowledge Graph (ORKG) [62]. This knowledge graph currently includes more than 170,000 resources, providing detailed information about the research contributions found in almost 15,000 scholarly articles across 709 research topics.

SciQA is a combination of manually curated and automatically generated question-query pairs. In detail, 100 pairs were manually crafted, and from these, the authors identified eight distinct question templates. Leveraging these templates, an additional 2,465 pairs were automatically generated using the GPT-3 model and subsequently verified by human experts [14]. An illustration of a manually generated pair (natural language question and SPARQL query) is presented in Example 1.

Example 1 Question: Which countries are considered in the papers about geopolitics?

```
SELECT DISTINCT ?location
WHERE { ?_ orkgp:compareContribution
        [ orkgp:P32
            [ rdfs:label ?label ];
        orkgp:P5049 ?location ]
    FILTER(REGEX(STR(?label), "geopolitics"))
}
```

The dataset in the benchmark is partitioned into three segments: 70% for training (1795 samples), 10% for validation (257 samples), and 20% for testing (513 samples). Table 1 presents pertinent statistics for each set. It is worth noting the imbalanced distribution of template frequencies across the datasets. The reader can refer to [4] for additional information regarding the SciQA dataset and its generation.

[5]SciQA dataset - https://huggingface.co/datasets/orkg/SciQA

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Table 1. We showcase various metrics for each template T_i across the training, validation, and test splits of the SciQA dataset, moving from left to right. These metrics include the count of questions crafted by humans, the count of questions generated automatically, the percentage of questions produced by humans (H), and the percentage of questions generated automatically.

	#Hum. Gen.	**#Aut. Gen.**	**H**	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**	**T7**	**T8**
Training	66	1729	3.7%	13.5%	13.3%	12.9%	13.3%	18.1%	2.7%	22.3%	0.3%
Validation	13	244	5.1%	14.0%	15.6%	14.8%	9.7%	16.3%	3.5%	21.0%	0%
Test	21	492	4.1%	12.5%	12.7%	14.2%	15.4%	18.5%	2.9%	19.5%	0.2%

3.2 Approaches

In our experiments, we considered six LLMs: i) T5-base[6][56], ii) GPT-2-large[7][55], iii) Dolly-v2-3b[8][22], iv) Mistral-7B-v0.1[9][32], v) Llama3.1-8B[10] and vi) GPT-3.5 Turbo[11][14].

These models utilize the transformer architecture, incorporating the attention mechanism to comprehend the relationships between words and sentences [67]. Our selection of these models was driven by their substantial diversity in both dimension (ranging from 220 million to 175 billion parameters) and structure (including both encoder-decoder and decoder-only configurations). This array of models allows us to assess a range of options in the benchmark. Here follows a brief overview of all the employed models, arranged according to their number of parameters.

The **T5-base** model adopts an encoder-decoder architecture comprising 220 million parameters and a context capacity of 512 tokens. Trained on the extensive C4 dataset[12], which encompasses approximately 750 GB of web pages, the model undergoes training through a mask-filling task, involving the random masking of 15% of the tokens. The **GPT-2-large** model employs a decoder-only structure, boasting 774 million parameters. With a maximum token capacity of 1024 covering both input and output, it underwent training on the WebText dataset[13], a compilation of around 40 GB of text extracted from 8 million web pages. The **Dolly-v2-3b** model utilizes a decoder-only design with 2.8 billion parameters and a token context capacity of 2048. Developed by Databricks, this model was fine-tuned based on EleutherAI's Pythia-12b[14]. To be more specific, Dolly-v2-3b underwent the fine-tuning process using a dataset containing approximately 15,000 instruction-response pairs. **Mistral-7B-v0.1** is an LLM with 7 billion parameters. It is equipped with grouped-query attention (GQA) for accelerated inference and incorporates sliding window attention (SWA) to efficiently handle sequences of arbitrary length at a reduced inference cost. **Llama3.1-8B** is an LLM with 8 billion parameters, built on a transformer-based architecture optimized for efficiency and scalability. It incorporates grouped-query attention (GQA) for accelerated inference and an enhanced positional encoding mechanism to improve long-context understanding. The model supports a context length of 128k tokens and has been trained on over 15 trillion tokens, making it highly versatile for multilingual language understanding and generation. Finally, the **GPT-3.5 Turbo** model adopts a decoder-only

[6]T5-base - https://huggingface.co/t5-base
[7]GPT-2-large - https://huggingface.co/gpt2-large
[8]Dolly-v2-3b - https://huggingface.co/databricks/dolly-v2-3b
[9]Mistral-7B-v0.1 - https://huggingface.co/mistralai/Mistral-7B-v0.1
[10]Llama3.1-8B -https://huggingface.co/meta-llama/Llama-3.1-8B
[11]GPT-3.5 Turbo - https://platform.openai.com/docs/models/gpt-3-5
[12]C4 dataset - https://huggingface.co/datasets/c4
[13]WebText dataset - https://paperswithcode.com/dataset/webtext
[14]Pythia-12b - https://huggingface.co/EleutherAI/pythia-12b

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structure comprising 175 billion parameters and accommodating a context of up to 16k tokens. This model extends the architecture and pre-training data of GPT-3, undergoing training on an extensive dataset of around 45 terabytes of text drawn from diverse sources such as books, websites, social media, and news articles. GPT-3.5 Turbo then underwent additional fine-tuning, focusing on data and tasks relevant to chat applications, including dialogues, question-and-answer sessions, command-response interactions, as well as code generation and execution.

4 METHODOLOGY

This section presents a thorough description of the experiments, including the various model variations, optimization strategies, experimental configuration, and the metrics employed.

4.1 Experiments

Our study focused on three optimization approaches for LLMs: fine-tuning (FT), zero-shot learning (ZSL), and few-shot learning (FSL). In the fine-tuning process, the model underwent refinement on the training set before its application to the test set. For zero-shot learning, the test set was directly processed using a basic prompt without any accompanying examples. Specifically, various prompt types were experimented with for each model, and the most effective prompt for each model was selected based on performance. Lastly, in few-shot learning, the prompt was enriched by incorporating a set of S samples extracted from the training set. When evaluating a question from the test set, we employed eight methods from the existing literature to identify the most pertinent samples for each question.

- **1) Random**: S samples are randomly selected for each element in the test set.
- **2) Similarity** [47]. The samples are ordered in descending fashion according to their semantic similarity with the question linked to the respective test set element, and the highest-ranked S samples are selected.
- **Diversity of template** [39, 44].
  - **3)** *Test A* (All diverse templates): The samples are arranged in a descending sequence based on their semantic similarity to the question, and the top S samples will be selected, ensuring that they represent different templates.
  - **4)** *Test B* (Same template for all): The samples are arranged in a descending sequence based on their semantic similarity to the question. Following this, the top S samples sharing the same template as the first sample are selected.
  - **5)** *DPPs* (Determinantal Point Processes [38]): The initial sample chosen is the one displaying the highest semantic similarity with the natural language question. The other six samples are selected based on having the lowest semantic similarity not only to each other but also to the first sample. The utilization of the DPPs approach was limited to seven samples to enhance coverage, and it was exclusively implemented for Dolly-v2-3B, Mistral-7B-v0.1, and GPT-3.5 Turbo, as these were the only models capable of accommodating prompts of such extended length.
- **Entropy** [30]. We applied three different techniques derived from [30]. For each template, the process begins by selecting the 8 shortest samples (in terms of string size) from the training set, and then:
  - **6)** *Same_Templ*: We start by identifying the most similar question in the training set and determining its template. Following this, we select the sample from the set of 8 samples that employs the same template.
  - **7)** *Ran_Templ*: We randomly choose a sample from the set of eight samples;
  - **8)** *Low_Perp*: We generate a SPARQL query for each of the eight input templates and subsequently calculate the perplexity for each query. The sample with the lowest perplexity is then selected.

We performed the experiments using six models: T5-base, GPT-2, Dolly-v2-3b, Mistral-7B-v0.1, Llama3.1-8B, and GPT-3.5-Turbo. These models were selected based on their performance on similar tasks, reputation within

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the research community, and diversity in parameter sizes, ensuring a comprehensive evaluation while staying within budget constraints. In the following, we describe them in detail.

4.1.1 T5-base model. The T5-base model exclusively underwent fine-tuning, as both zero-shot learning (ZSL) and few-shot learning (FSL) were deemed impractical due to the constrained prompt size and the model's pre-training on specific tasks such as translating from English to German or summarizing text. Consequently, when tasked with English to SPARQL translation, the model exhibited a tendency to offer a German translation of the question instead. During the fine-tuning (FT) process, the T5 model was tasked with translating a given English natural language question into a SPARQL query. The designated prefix for this task was *"Translate English to SPARQL:"*.

The fine-tuning process involved utilizing 1,795 labeled samples from the training set across 20 epochs. Each sample comprised a brief prefix and a natural language question, serving as the input request to the model, and was associated with the corresponding SPARQL query, forming the expected response. Example 2 provides an illustration of the data type employed in the fine-tuning process of the T5-base model.

Example 2 Type of data employed during the fine-tuning phase of the T5-base model.

Request:
"Translate English to SPARQL: Which model has achieved the highest Accuracy score on the ARC (Challenge) benchmark dataset?"

Excpected output:

```
SELECT DISTINCT ?model ?model_lbl
WHERE {
    ?metric a orkgc:Metric ;
            rdfs:label ?metric_lbl .
    FILTER str?metric_lbl = "Accuracy"
    {
        SELECT ?model ?model_lbl
        WHERE {
            ?dataset a orkgc:Dataset ;
                     rdfs:label ?dataset_lbl .
            FILTER str?dataset_lbl = "ARC Challenge"
            ?benchmark orkgp:HAS_DATASET ?dataset ;
                       orkgp:HAS_EVALUATION ?eval .
            ?eval orkgp:HAS_VALUE ?value ;
                  orkgp:HAS_METRIC ?metric .
            ?cont orkgp:HAS_BENCHMARK ?benchmark ;
                  orkgp:HAS_MODEL ?model .
            ?model rdfs:label ?model_lbl .
        }
        ORDER BY DESC?value
        LIMIT 1
    }
}
```

4.1.2 GPT-2-large model. The principal constraint of the GPT-2 model was its limited context size, imposing restrictions on the utilization of extended prompts and multiple samples in few-shot learning (FSL). Despite this limitation, we effectively executed a comprehensive set of experiments. In particular, we conducted the following experiments:

- ZSL, using the prompt: *"input (English text): [nl question] \n output (Sparql query):"*.
- FSL with 1, 2, and 3 samples, constrained by the 1024 token context limitation, which precluded the inclusion of additional samples. In the subsequent description, we present an example using 1-shot learning:
  "input (English text): [the sample question in natural language]"
  "output (Sparql query): [the sample SPARQL query]"
  "input (English text): [the question in natural language of the underlying test set sample]"
  "output (SPARQL query):"

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We explored all the methods outlined in the preceding section, including similarity, diversity, and entropy, to choose relevant samples.
- FT: The GPT-2-large model underwent fine-tuning using a text file containing all question and query pairs from the training set. The special token < |endoftext| > was inserted before each pair, acting as a signal for the model to cease output generation. The fine-tuning process spanned 20 epochs.

4.1.3 Dolly-v2-3b model. We performed the complete set of experiments including fine-tuning (FT), zero-shot learning (ZSL), and few-shot learning (FSL). Specifically, we conducted the following experiments:
- ZSL, using the prompt: *"Translate to a SPARQL query the following English question:"*.
- FSL: We employed 1, 3, 5, and 7 examples for few-shot learning. The model's token limit of 2,048 constrained our input to a maximum of seven samples. Utilizing the same prompt format as with the GPT-2 model, we consistently applied the entire spectrum of previously discussed metrics for the purpose of sample selection in testing.
- FT: Dolly-v2-3b underwent fine-tuning using the pairs of natural language questions and SPARQL queries from the training set and using 5 epochs.

4.1.4 Mistral-7B-v0.1 model. With Mistral-7B-v0.1, we carried out a comprehensive set of experiments encompassing fine-tuning (FT), zero-shot learning (ZSL), and few-shot learning (FSL). The following descriptions outline the conducted experiments:
- ZSL, using the prompt: *"Translate to a SPARQL query the following English text:"*.
- FSL: Few-shot experiments were conducted using 1, 3, 5, and 7 samples. The samples were provided using the same template as utilized for GPT-2 and Dolly-v2-3b models.
- FT: Mistral-7B-v0.1 underwent fine-tuning using the 1795 pairs of natural language questions and SPARQL queries from the training set, with training carried out over 5 epochs.

4.1.5 Llama3.1-8B model. The Llama3.1-8B model was exclusively fine-tuned, as it was introduced at a later stage of the experimentation phase, and fine-tuning had already proven to be the most effective approach for the other models.

During the fine-tuning process, the Llama3.1-8B model was trained to translate English natural language questions into SPARQL queries using a structured prompt format. The fine-tuning prompt followed this template: *"Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.*
Instruction: Given an English text describing an information need, generate a SPARQL query to retrieve relevant data from the [Open Research Knowledge Graph (ORKG) or DBLP Knowledge Graph]
Input: [the sample question in natural language]
Response: [the sample SPARQL query]" followed by the *EOS_TOKEN* from the tokenizer. The fine-tuning process was performed over 10 epochs.

4.1.6 GPT-3.5-Turbo. Similar to Dolly, the potential resource intensity associated with fine-tuning, given the model's large size, led us to focus on zero-shot learning (ZSL) and few-shot learning (FSL). In summary, the following experiments were conducted:
- ZSL, using the prompt: *"What is the SPARQL query for:"*.
- FSL: To maintain comparability with the other models, we conducted experiments using 1, 3, 5, and 7 samples. The entire set of methodologies for sample selection was applied, with the exception of the Entropy - *Low_Perp* methodology. This exclusion was due to the impracticality of computing perplexity with the OpenAI API. The samples were provided using the following template:
"input (English text): [the sample question in natural language]"

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"output (SPARQL query): [the sample SPARQL query]"
"What is the SPARQL query for: [the question in natural language]"

4.2 Experimental Setup

We employed a comprehensive set of metrics to assess the performance of the models over SciQA, including Precision, Recall, F1 Score, Bleu 4, Bleu Cumulative, Rouge 1, and Rouge 2.

Precision is calculated as the ratio of overlapping tokens between the predicted output and the ground truth, divided by the total number of tokens in the predicted output. Recall is computed by dividing the number of overlapping tokens by the total number of tokens in the ground truth. The F1-score is the harmonic mean of precision and recall. The Blue (Bilingual Evaluation Understudy) score measures the quality of predicted text by comparing it to a set of reference texts. It ranges from 0 to 1, with higher scores indicating better prediction quality (scores above 0.3 are typically considered good). Specifically, the Blue 4 score evaluates how many 4-grams in the candidate translation match those in the ground truth, focusing primarily on precision. Cumulative Blue (Blue C) aggregates n-gram accuracy across different n-values and computes the geometric mean. Finally, Rouge-n assesses the overlap of n-grams between the model-generated text and the ground truth. In particular, we have used Rouge-1 and Rouge-2.

For a detailed overview of these metrics, we refer the reader to the work of Chauhan and Daniel [18]. The experimental setup for the approaches is detailed as follows:

T5-base was initially fine-tuned using the HuggingFace *Seq2SeqTrainer*[15] with specific parameters: learning_rate=2e-5, weight_decay=0.01, fp16=True. Subsequently, the *generate* method of the *AutoModelForSeq2SeqLM*[16] class from the Transformers[17] library was utilized, incorporating the settings max_new_tokens=512, do_sample=True, top_k=30, top_p=0.95. GPT-2-large underwent fine-tuning using the HuggingFace *Trainer*[18] class with default parameters. GPT-2-large was utilized through the *pipeline*[19] abstraction from the Transformers[17] library employing parameters such as task=text-generation, max_new_tokens=400, return_full_text=False. Dolly-v2-3b was also employed using the *pipeline* abstraction from the Transformers[17] library and setting parameters like torch_dtype=torch.bfloat16, trust_remote_code=True.

Mistral-7B-v0.1 was employed in conjunction with the *bitsandbytes* library, utilizing parameters such as load_in_4bit=True, bnb_4bit_use_double_quant=True, bnb_4bit_quant_type="nf4", and bnb_4bit_compute_dtype=torch.bfloat16. Additionally, the *AutoTokenizer* and *AutoModelForCausalLM* classes from the Transformers library were utilized. For the generation process, the *generate* method was employed with max_new_tokens=1000 and do_sample=True.

In the fine-tuning process, QLORA[20] (a library used for the fine-tuning of Mistral-7B-v0.1 and Dolly-v2-3b) was utilized with the following configuration: r=8, lora_alpha=16, bias="none", lora_dropout=0.05, task_type="CAUSAL_LM".

Llama3.1-8B was loaded using the *FastLanguageModel* class from the *unsloth* library[21], with a maximum sequence length of 2048 tokens. The data type (*dtype*) was automatically detected based on the hardware, while 4-bit quantization (*load_in_4bit=True*) was enabled to reduce memory usage. The model and its corresponding tokenizer were loaded using the *from_pretrained* method. The model was fine-tuned using the *SFTTrainer* class from the *trl* library, leveraging the *TrainingArguments* module from Transformers. The training process utilized a

[15]Seq2SeqTrainer - https://huggingface.co/docs/transformers/v4.35.1/en/main_classes/trainer#transformers.Seq2SeqTrainer
[16]AutoModelForSeq2SeqLM - https://huggingface.co/docs/transformers/v4.35.1/en/model_doc/auto#transformers.AutoModelForSeq2SeqLM
[17]Transformers - https://huggingface.co/docs/transformers/v4.35.1/en/index
[18]Trainer - https://huggingface.co/docs/transformers/v4.35.1/en/main_classes/trainer
[19]Pipeline class - https://huggingface.co/docs/transformers/v4.35.1/en/main_classes/pipelines
[20]https://github.com/artidoro/qlora
[21]https://unsloth.ai/

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per-device batch size of 2, gradient accumulation steps of 4, and a learning rate of 2e-4. Mixed precision training was enabled, with *fp16* or *bf16* automatically selected based on hardware support. Optimization was performed using the *adamw_8bit* optimizer, with a weight decay of 0.01 and a linear learning rate scheduler. The model was fine-tuned for 10 epochs, with logging every 500 steps. For the generation process, the model was loaded using the *FastLanguageModel* class with a maximum sequence length of 2048 tokens, utilizing the specified settings of *dtype=None* and *load_in_4bit=True*. It was prepared for inference using *FastLanguageModel.for_inference*. The *generate* method was employed with *max_new_tokens=800* and *use_cache=True*, and the final outputs were decoded using *tokenizer.batch_decode*.

Finally, GPT-3.5 Turbo was accessed through the OPENAI API[22] using the *create* method of the *ChatCompletion*[23] class from the *openai*[24] Python library. Parameters were configured as follows: system role: *"You are a translator from natural language to SPARQL using the provided examples."*, temperature=0.5, top_p=0.95, frequency_penalty=0, presence_penalty=0, stop=None, timeout=30.

To ensure reproducibility, we have made available a repository containing both the codebase and the prompts used in the experiments[25].

5 RESULTS AND DISCUSSION

This section presents the results of the evaluation on SciQA (Section 5.1), followed by an error analysis aimed at identifying recurring error patterns (Section 5.2).

5.1 Performance on SciQA

The performance of the six models according to the full set of the metrics outlined in Section 4.2, are reported in Table 2 (T5), Table 3 (GPT-2), Table 4 (Dolly), Table 5 (Mistral-7B-v0.1), Table 6 (GPT-3.5) and Table 7 (Llama3.1-8B). Table 8 offers a comparative overview of all the configurations focusing on the F1-scores and exact matches.

Table 2. T5-base model - Fine-tuned.

Strat.	Prec.	Rec.	F1 Score	Bleu 4	Bleu C	Rouge 1	Rouge 2	Exact matches
FT	0.9767	0.9760	0.9751	0.9597	0.9631	0.9790	0.9683	483

The fine-tuned Mistral-7B-v0.1 model led with the highest F1 score (97.69%), narrowly outperforming the fine-tuned Llama3.1-8B (97.65%) and the fine-tuned T5 model (97.51%). It was followed by GPT-3.5 using a seven-sample few-shot approach with samples selected based on similarity (97.36%), the fine-tuned GPT-2 (96.69%), and the fine-tuned Dolly (96.58%).

When considering exact matches, Llama3.1-8B achieved the best score (95.9%, with 492 out of 513 exact matches), while T5 and Dolly ranked second (94.1%, 483 out of 513). Close behind is the fine-tuned Mistral-7B-v0.1 (93.6%), followed by GPT-3.5 employing the similarity-based 7-sample few-shot approach (92.6%), and the fine-tuned GPT-2 (83.8%). Manual analysis suggests that the strong performance of T5 and Dolly may be related to their more constrained 'creativity' compared to bigger models. The larger models sometimes generate alternative predicates or interpret semantics in a more inventive manner. In Section 5.2, we will conduct a detailed analysis of these error types.

[22]OPENAI API - https://openai.com/blog/openai-api
[23]ChatCompletion - https://platform.openai.com/docs/guides/text-generation/chat-completions-api
[24]OpenAI Libraries - https://platform.openai.com/docs/libraries
[25]Codebase and prompts - https://github.com/paper-support-materials/Analysis-of-the-SciQA-Benchmark

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Table 3. GPT2-large model - Performance based on the strategy (Strat.) and the criteria governing Few-Shot Learning (FSL) sample selection (C): Similarity, Entropy, Random, Diversity. *S* denotes the number of samples. The best results for each metric are highlighted in bold.

Strat.	C	Test name	S	Prec.	Rec.	F1 Score	Bleu 4	Bleu C	Rouge 1	Rouge 2	Exact matches
FT				**0.9693**	**0.9669**	**0.9669**	**0.9462**	**0.9504**	**0.9708**	**0.9580**	**430**
ZSL				0.0464	0.1579	0.0653	0.0004	0.0009	0.0932	0.0087	0
FSL	Simil.		1	0.1477	0.4877	0.2076	0.0685	0.0822	0.2535	0.1640	0
			2	0.2054	0.5112	0.2719	0.1205	0.1355	0.3237	0.2454	1
			3	0.2366	0.5692	0.3107	0.1502	0.1670	0.3767	0.3077	2
	Entro.	Same_Templ	1	0.1567	0.3903	0.2029	0.0541	0.0696	0.2383	0.1373	0
		Ran_Templ	1	0.1105	0.2778	0.1421	0.0180	0.0280	0.1798	0.0675	0
		Low_Perp	1	0.2408	0.4360	0.2788	0.0924	0.1263	0.3427	0.2302	0
	Rand.		1	0.1119	0.3001	0.1499	0.0191	0.02950	0.1826	0.0692	0
			2	0.1230	0.2968	0.1580	0.0255	0.0372	0.1976	0.0807	0
			3	0.1336	0.3195	0.1730	0.0303	0.0449	0.2287	0.1003	0
	Diver.	Test A	3	0.1680	0.4422	0.2215	0.0519	0.0721	0.2879	0.1573	0
		Test B	3	0.2205	0.5636	0.2988	0.1496	0.1635	0.3555	0.2926	1

When examining the additional metrics presented in Section 4.2 and outlined in Tables 2-7, the trend is consistent. The leading four models across Rouge 1, Rouge 2, Bleu 4, and Bleu C metrics remain the fine-tuned Llama3.1-8B, the fine-tuned Mistral-7B-v0.1, the fine-tuned T5, and GPT 3.5 utilizing 7-sample few-shot learning. Specifically, Mistral-7B-v0.1 stood out as the best model according to the Rouge-1 metric, whereas T5 showed superior performance in Rouge-2. Meanwhile, Llama3.1-8B led in Bleu-4 and Bleu-C, whereas GPT-3.5 consistently ranked fourth.

The experiments offered many significant insights.

An initial observation relates to the difficulty of the SciQA benchmark. This benchmark posed a significant challenge for all models under zero-shot learning conditions, with no exact matches and F1 scores below 26%. These poor performances can be attributed to two key factors. First, LLMs are not specifically optimized for translating natural language into SPARQL queries. Second, they lack an understanding of the underlying data schema, even though some relationships and entity types from Wikidata may be present in their training data. However, performance improves significantly when fine-tuning is applied, allowing the model to learn the schema, and when few-shot learning is used, offering examples of schema usage through similar queries. For instance, the fine-tuned version of Llama3.1-8B achieved outstanding results, attaining nearly 96% exact matches and exceeding a 97% F1 score. This success can be partly attributed to the regularities within the benchmark, which fine-tuned models were able to detect and replicate. To push the boundaries of this field further, it is essential to develop more diverse and challenging datasets that encompass a wider variety of realistic query types, while actively involving human users in the dataset creation process.

A second noteworthy outcome is that several relatively small models (Mistral-7B-v0.1, T5, and Llama3.1-8B), when fine-tuned with relevant data, outperformed the larger GPT-3.5 model. This achievement is especially notable for T5, which, with only 220M parameters, outperformed several larger counterparts including GPT-2, GPT-3.5, and Dolly. This brings further evidence to the notion that fine-tuning on well-curated datasets enables

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Table 4. Dolly-v2-3b model - Performance based on the strategy (Strat.) and the criteria governing Few-Shot Learning (FSL) sample selection (C): Similarity, Entropy, Random, Diversity. *S* denotes the number of samples. The best results for each metric are highlighted in bold.

Strat.	C	Test name	S	Prec.	Rec.	F1 Score	Bleu 4	Bleu C	Rouge 1	Rouge 2	Exact matches
FT				**0.9661**	**0.9722**	**0.9658**	**0.9512**	**0.9530**	**0.9722**	**0.9601**	**483**
ZSL				0.1911	0.0993	0.1087	0.0033	0.0062	0.1734	0.0358	0
FSL	Similarity		1	0.8742	0.9088	0.8792	0.7728	0.8015	0.8861	0.8494	167
			3	0.7979	0.9163	0.8304	0.7204	0.7432	0.8775	0.8470	182
			5	0.7845	0.9238	0.8242	0.7102	0.7322	0.8711	0.8407	180
			7	0.7681	0.9057	0.8052	0.6911	0.7113	0.8515	0.8215	181
	Entr.	Same_Templ	1	0.4866	0.8089	0.5734	0.3662	0.4005	0.5741	0.5474	2
		Ran_Templ	1	0.3789	0.6107	0.4402	0.1604	0.2155	0.4811	0.3288	0
		Low_Perp	1	0.5854	0.8647	0.6757	0.4499	0.5013	0.7262	0.6201	1
	Random		1	0.5976	0.5973	0.5659	0.2799	0.3456	0.6269	0.4559	27
			3	0.5830	0.6676	0.5900	0.3484	0.4038	0.6435	0.5131	31
			5	0.6075	0.7147	0.6242	0.3934	0.4450	0.6847	0.5575	51
			7	0.6358	0.7423	0.6576	0.4460	0.4947	0.7153	0.6001	69
	Diversity	Test A	3	0.6621	0.8240	0.7000	0.4587	0.5138	0.7766	0.6565	43
			5	0.6350	0.8151	0.6825	0.4329	0.4912	0.7702	0.6327	39
			7	0.6316	0.7941	0.6729	0.4246	0.4836	0.7529	0.6116	46
		Test B	3	0.7623	0.9068	0.8025	0.6926	0.7139	0.8533	0.8254	171
			5	0.7793	0.9223	0.8181	0.7148	0.7346	0.8647	0.8411	201
			7	0.7961	0.9122	0.8261	0.7279	0.7456	0.8647	0.8398	212
		DPPs	7	0.4178	0.4234	0.3912	0.2764	0.2868	0.4289	0.3387	54

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the development of efficient low-resource models that can deliver performance on par with much more resource-intensive solutions. We further evaluate the performance of these models in Section 6, showing that T5 is particularly well-suited for this task when supplied with contextual information about entities and relationships. This approach could enable the development of a lightweight and cost-effective solution for retrieving relevant data from scientific knowledge graphs and feeding it to larger LLMs. Such a method has the potential to support a new generation of question-answering systems capable of producing higher-quality responses grounded in structured, verifiable information.

A third important finding concerns the metrics used for retrieving relevant examples in few-shot learning. The experiments demonstrated that simple semantic similarity consistently outperformed other methods across all models. Notably, several more complex metrics recently introduced in the literature [30, 39, 44] did not perform as well on this benchmark. Additionally, we noted that the response of various models to changes in sample size varied. For instance, GPT 3.5 demonstrated enhanced results with an increase in sample size when employing a similarity-based selection method, whereas Dolly's performance deteriorated, indicating that a larger number of examples might overwhelm this model. These observations are worth further exploration in future research.

Table 5. Mistral-7B V0.1 model - Performance based on the strategy (Strat.) and the criteria governing Few-Shot Learning (FSL) sample selection (C): Similarity, Entropy, Random, Diversity. *S* denotes the number of samples. The best results for each metric are highlighted in bold.

Strat.	C	Test name	S	Prec.	Rec.	F1 Score	Bleu 4	Bleu C	Rouge 1	Rouge 2	Exact matches
FT				**0.9772**	**0.9765**	**0.9769**	**0.9592**	**0.9622**	**0.9798**	**0.9682**	**480**
ZSL				0.3266	0.1786	0.2043	0.0033	0.0068	0.3132	0.0625	0
FSL	Simil.		1	0.8986	0.8538	0.8529	0.7439	0.7709	0.8872	0.8198	259
			3	0.9462	0.9442	0.9409	0.8901	0.9037	0.9543	0.9240	382
			5	0.9482	0.9396	0.9386	0.8952	0.9051	0.9517	0.9262	403
			7	0.9492	0.9514	0.9470	0.9091	0.9188	0.9562	0.9340	409
	Entro.	Same_Templ	1	0.8260	0.7849	0.7765	0.6198	0.6582	0.8243	0.7235	166
		Ran_Templ	1	0.5765	0.4695	0.4711	0.1553	0.2139	0.5829	0.3248	17
		Low_Perp	1	0.8442	0.8404	0.8251	0.6721	0.7108	0.8723	0.7751	189
	Rand.		1	0.5172	0.5542	0.5055	0.1871	0.2502	0.6086	0.3619	28
			3	0.6816	0.7341	0.6835	0.4245	0.4895	0.7591	0.5924	75
			5	0.7440	0.7736	0.7342	0.5116	0.5672	0.7926	0.6601	115
			7	0.7816	0.8174	0.7799	0.5852	0.6357	0.8271	0.7209	135
	Diversity	Test A	3	0.8880	0.8629	0.8589	0.7067	0.7475	0.9058	0.8253	154
			5	0.8678	0.8540	0.8447	0.6747	0.7225	0.8957	0.8048	120
			7	0.8203	0.8216	0.8033	0.5871	0.6463	0.8702	0.7476	84
		Test B	3	0.9336	0.9156	0.9179	0.8391	0.8584	0.9380	0.8889	368
			5	0.9281	0.9115	0.9095	0.8404	0.8572	0.9300	0.8862	378
			7	0.9318	0.9056	0.9087	0.8407	0.8558	0.9273	0.8831	389
		DPPs	7	0.7616	0.7275	0.7277	0.4806	0.5414	0.7860	0.6329	84

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5.2 Error Analysis

We carried out an extensive review of queries that were classified as incorrect due to their deviation from the benchmark responses. We focused on the error patterns emerging from the top models: Mistral-7B-v0.1 (33 errors), T5 (30 errors), GPT-3.5 (38 errors), and Llama3.1-8B (21 errors).

We examined the incorrect queries generated by three models and identified five categories of common mistakes. Table 9 outlines the definitions of these categories and reports the respective percentages of incorrect queries for Mistral-7B-v0.1, T5, GPT-3.5, and Llama3.1-8B. It is important to note that a single query may be mapped to multiple categories (e.g., it can contain both a misspelled entity and a wrong entity type), so the combined percentage for all categories is greater than 100%. In the following, we will analyze each error category and the performance of the three models concerning each.

The most frequent error is the generation of an incorrect predicate (e.g., using *orkgp:HAS_METHOD* in place of the correct *orkgp:P7046*). The *wrong predicate* error type affects all models, though to different degrees. Among the models, T5 is the least susceptible to this issue (36.6%), with approximately one-third of its erroneous queries falling into this category. Mistral-7B-v0.1 exhibits this error in over half of its erroneous outputs (57.6%), while GPT-3.5 (76.3%) and Llama3.1-8B (85.7%) show an even higher rate of such mistakes. This pattern highlights the challenge LLMs face in accurately using specialized predicate vocabularies, which can significantly differ from standard verb usage or more popular predicates used by large knowledge graphs, even after being fine-tuned

Table 6. GPT-3.5-Turbo model - Performance based on the strategy (Strat.) and the criteria governing Few-Shot Learning (FSL) sample selection (C): Similarity, Entropy, Random, Diversity. *S* denotes the number of samples. The best results for each metric are highlighted in bold.

Strat.	C	Test name	S	Prec.	Rec.	F1 Score	Bleu 4	Bleu C	Rouge 1	Rouge 2	Exact matches
ZSL				0.4963	0.1931	0.2632	0.0039	0.0088	0.4010	0.1019	0
FSL	Similarity		1	0.9509	0.9305	0.9368	0.8655	0.8833	0.9505	0.9137	356
			3	0.9730	0.9635	0.9667	0.9378	0.9439	0.9750	0.9558	451
			5	0.9759	0.9685	0.9709	0.9481	0.9521	0.9772	0.9609	464
			7	**0.9768**	**0.9727**	**0.9736**	**0.9534**	**0.9571**	**0.9788**	**0.9638**	**475**
	Ent.	Same_Templ	1	0.9303	0.8781	0.8988	0.7759	0.8119	0.9252	0.8631	205
		Ran_Templ	1	0.8268	0.6345	0.7016	0.3222	0.4070	0.7946	0.5651	26
	Random		1	0.8162	0.6962	0.7362	0.3954	0.4828	0.8194	0.6107	45
			3	0.8707	0.8052	0.8259	0.5914	0.6558	0.8817	0.7543	113
			5	0.9024	0.8499	0.8675	0.6813	0.7343	0.9073	0.8141	165
			7	0.9226	0.8726	0.8905	0.7356	0.7799	0.9227	0.8476	189
	Diversity	Test A	3	0.9693	0.9156	0.9378	0.8613	0.8784	0.9619	0.9232	315
			5	0.9730	0.9209	0.9428	0.8699	0.8875	0.9642	0.9290	328
			7	0.9688	0.9158	0.9375	0.8599	0.8783	0.9610	0.9230	313
		Test B	3	0.9678	0.9492	0.9561	0.9057	0.9181	0.9672	0.9391	412
			5	0.9683	0.9502	0.9566	0.9076	0.9192	0.9681	0.9401	417
			7	0.9673	0.9501	0.9562	0.9061	0.9180	0.9671	0.9386	422
		DPPs	7	0.9623	0.9355	0.9452	0.8833	0.8982	0.9610	0.9261	389

Table 7. Llama3.1-8B model - Fine-tuned.

Strat.	Prec.	Rec.	F1 Score	Bleu 4	Bleu C	Rouge 1	Rouge 2	Exact matches
FT	0.9793	0.9765	0.9765	0.9613	0.9637	0.9796	0.9681	492

with relevant data. Example 5 shows an exemplary case in which GPT-3.5 hallucinates a fictitious predicate (*orkgp:P2067*), probably influenced by the Wikidata predicate *Property:P2067*.

The second most common mistake identified is the semantic error, occurring when the model misinterprets the original question's meaning and generates a query with entirely different semantics. Similar to the issue of incorrect predicates, T5 is the least prone to this error (26.6%), with Llama3.1-8B (42.9%), GPT-3.5 (57.8%), and Mistral-7B-v0.1 (69.7%) much more likely to make such mistakes. It is noteworthy that T5, despite being the smallest model, demonstrates superior performance in handling the complex challenge of understanding the semantics of natural language queries.

The next most notable error type is the *misspelled entity* category, where models alter entity names, frequently substituting them with synonyms (e.g, "robotic navigation" instead of "robot navigation"). An illustrative instance is provided in Example 3, where the entity *linear-chain CRFs* was modified to *label-chain CRFs*. Remarkably, T5 and GPT-3.5 display almost identical tendencies in this regard, with error rates of 60.0% and 60.5%, respectively.

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Table 8. Summary of F1-scores and exact matches (in parentheses), categorized by different strategies (Strat.) and criteria (C) such as Similarity, Entropy, Random, and Diversity. *S* represents the number of samples, and the best results for each model are highlighted in bold.

Strat.	C	Test name	S	T5-base	GPT2-large	Dolly-v2-3b	Mistral-7B	GPT-3.5-turbo	Llama3.1-8B
FT				**0.9751 (483)**	**0.9669 (430)**	**0.9658 (483)**	**0.9769 (480)**		**0.9765 (492)**
ZSL					0.0653 (0)	0.1087 (0)	0.2043 (0)	0.2632 (0)	
FSL	Similarity		1		0.2718 (0)	0.8792 (167)	0.8529 (259)	0.9368 (356)	
			3		0.4051 (2)	0.8304 (182)	0.9409 (382)	0.9667 (451)	
			5			0.8242 (180)	0.9386 (403)	0.9709 (464)	
			7			0.8052 (181)	0.9470 (409)	**0.9736 (475)**	
	Ent.	Same_Templ	1		0.2029 (0)	0.5734 (2)	0.7765 (166)	0.8988 (205)	
		Ran_Templ	1		0.1421 (0)	0.4402 (0)	0.4711 (17)	0.7016 (26)	
		Low_Perp	1		0.2788 (0)	0.6757 (1)	0.8251 (189)		
	Random		1		0.2005 (0)	0.5659 (27)	0.5055 (28)	0.7362 (45)	
			3		0.2187 (0)	0.5900 (31)	0.6835 (75)	0.8259 (113)	
			5			0.6242 (51)	0.7342 (115)	0.8675 (165)	
			7			0.6576 (69)	0.7799 (135)	0.8905 (189)	
	Diversity	Test A	3		0.2215 (0)	0.7000 (43)	0.8589 (154)	0.9378 (315)	
			5			0.6525 (39)	0.8447 (120)	0.9428 (328)	
			7			0.6729 (46)	0.8033 (84)	0.9375 (313)	
		Test B	3		0.2988 (1)	0.8025 (171)	0.9179 (368)	0.9561 (412)	
			5			0.8181 (201)	0.9095 (378)	0.9566 (417)	
			7			0.8261 (212)	0.9087 (389)	0.9562 (422)	
		DPPs	7			0.3912 (54)	0.7277 (84)	0.9452 (389)	

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Llama3.1-8B shows a moderate performance in this category, with a misspelling rate of 42.9%. Mistral-7B-v0.1 (33.3%) is the clear winner of this category, demonstrating a much lower incidence of misspelled entities.

Regarding the *wrong entity type* category (e.g., the query searching an entity of type Energy Source instead of one of type Paper), Mistral-7B-v0.1 stands out as the model least affected by such errors (15.2%), in contrast to T5 (36.6%), GPT-3.5 (52.6%) and Llama3.1-8B (90.5%), where this issue is more prevalent. Notably, Llama3.1-8B exhibits this error in the vast majority of its incorrect queries. This suggests that Mistral-7B-v0.1 has a more accurate representation of the various object categories within a query, allowing it to more effectively distinguish between different types of entities.

Finally, the less frequent error type across the three top models is the *SPARQL syntactic error* category, which predominantly affects T5 (40.0%). While T5 often avoids other types of errors, it sometimes struggles to generate correct SPARQL syntax. Example 4 reports a typical case in which T5 generates a syntactically incorrect query. Interestingly, GPT-3.5 (5.2%), Mistral-7B-v0.1 (6.1%), and Llama3.1-8B (9.5%) are largely unaffected by this type of error. This may suggest that larger models have a better capability to adhere to the syntax of query languages, thereby less frequently producing incorrect code that cannot be interpreted by a SPARQL engine.

In conclusion, the four models exhibit distinct error patterns. T5, GPT-3.5, and Llama3.1-8B struggled with understanding the underlying ontological schema, resulting in several misclassified entities and incorrect entity types. In contrast, Mistral-7B-v0.1 performed significantly better in these areas, but exhibited a higher incidence of semantic misunderstandings.

Table 9. Error categories and their frequencies for Mistral-7B-v0.1, T5, GPT-3.5, and Llama3.1-8B.

Error Category	Category Definition	Mistral-7B-v0.1	T5	GPT-3.5	Llama3.1-8B	Average
(1) Wrong predicate	The query uses an incorrect predicate in the query.	57.6%	36.6%	76.3%	85.7%	64.1%
(2) Semantic error	The query does not reflect the meaning of the question.	69.7%	26.6%	57.8%	42.9%	49.3%
(3) Misspelled entity	The query misspells an entity name.	33.3 %	60.0%	60.5%	42.9%	49.2%
(4) Wrong entity type	The query uses an entity of a different type.	15.2%	36.6%	52.6%	90.5%	49.6%
(5) SPARQL syntactic error	The query cannot be syntactically parsed by a SPARQL compiler.	6.1%	40.0%	5.2%	9.5%	15.2%

Example 3 Example generated by T5 for the *misspelled entity* category. The original entity is reported in blue and in red is reported the misspelled one. Question: *List the code links in papers that use the linear-chain CRFs model in any benchmark?*

Original query:

```
SELECT DISTINCT ?code
WHERE {
    ?model a orkgc:Model ;
           rdfs:label ?model_lbl .
    FILTER (str(?model_lbl) = "linear-chain CRFs")
    ?benchmark orkgp:HAS_DATASET ?dataset .
    ?cont orkgp:HAS_BENCHMARK ?benchmark .
    ?cont orkgp:HAS_MODEL ?model ;
          orkgp:HAS_SOURCE_CODE ?code .
}
```

Generated output:

```
SELECT DISTINCT ?code
WHERE {
    ?model a orkgc:Model;
        rdfs:label ?model_lbl.
    FILTER (str( ?model_lbl) = "label-chain CRFs")
    ?benchmark orkgp:HAS_DATASET ?dataset.
    ?cont orkgp:HAS_BENCHMARK ?benchmark.
    ?cont orkgp:HAS_MODEL ?model;
        orkgp:HAS_SOURCE_CODE ?code.
}
```

Example 4 Example generated by T5 for the *SPARQL syntactic error* category. The non-matching section is highlighted in red. Question: *What is Raman spectroscopy?*

Original query:

```
SELECT ?definitions
WHERE {
    ?terms orkgp:P24009 ?definitions .
    ?terms rdfs:label ?terms_labels .
    FILTER regex(?terms_labels, "Raman spectroscopy")
}
```

Generated output:

```
SELECT ?rmdfs:label ?rmdfs:label
?rmdfs:label ?rmdfs:label ?rmdfs:label
?rmdfs:label ?rmdfs:label ?rmdfs:label
?rmdfs:label ?rmdfs:label ?rmdfs:label
?rmdfs:label. }
```

Crucially, the first type of error may be easier to address, for instance, by incorporating an additional entity recognition module. To investigate this further, the next section evaluates the same four models on the DBLP-QuAD benchmark. DBLP-QuAD is similar to SciQA, but it also provides relevant entities and relationships in the input, allowing us to assess whether supplying accurate entity information can help mitigate these errors.

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Example 5 Example generated by GPT-3.5 for the *wrong predicate* category. The hallucinated predicate is highlighted in red. Question: *What quantity of iron oxide was discovered on Elorza crater?*

Original query:

```
SELECT ?properties_values ?property_description
WHERE {
    ?papers rdf:type orkgc:Paper .
    ?papers rdfs:label ?papers_labels .
    ?papers orkgp:P31 ?contrib .
    ?contrib ?properties ?properties_values .
    ?properties rdfs:label ?properties_labels .
    ?properties orkgp:description ?property_description .
    FILTER regex(?properties_labels, "FeO")
    FILTER regex(?papers_labels, "Elorza crater", "i")
}
```

Generated output:

```
SELECT ?quantity
WHERE {
    ?crater rdf:type orkgc:Crater.
    ?crater rdfs:label "Elorza crater".
    ?crater orkgp:P31 ?iron_oxide_discoveries.
    ?iron_oxide_discoveries orkgp:P2067 ?quantity.
}
```

6 ADDITIONAL EXPERIMENTS ON DBLP-QUAD

In this section, we present additional experiments involving the four top-performing models identified in the previous evaluation. Section 6.1 examines their performance on DBLP-QuAD. In Section 6.2, we assess the models using additional execution-based metrics, offering further insights into their practical applicability. Section 6.3 introduces a retrieval-augmented ensemble method and shows that it achieves superior performance compared to each individual model on the DBLP-QuAD dataset. Section 6.4 explores transfer learning between SciQA and DBLP-QuAD by evaluating how models trained on one benchmark perform when applied to the other. Finally, Section 6.5 outlines the computational resources and associated costs of the experiments.

6.1 Performance on DBLP-QuAD

The DBLP-QuAD benchmark[26] [6] consists of 10,000 unique question-query pairs, which are split into training, validation, and test sets in a 7:1:2 ratio. It covers 13,348 entities and 11 predicates derived from the DBLP Knowledge Graph, which is a machine-readable version of DBLP. It features 10 types of queries, with 1,000 question-query pairs for each type, evenly distributed between author-centric and publication-centric queries. Furthermore, 2,350 of the questions are temporal in nature, requiring the evaluation of statistics over a defined time period, such as "in the last five years." A key distinction from the SciQA dataset is that DBLP-QuAD explicitly provides the entities and relationships referenced in the natural language query, enabling models to utilize this information for more accurate predictions.

We evaluated the top four models from our previous experiments using the DBLP-QuAD dataset. It is important to highlight that, in contrast to the original paper by Banerjee et al [6], introducing this benchmark, we fine-tuned the T5 model for 20 epochs instead of 5 and used a slightly modified prompt. These adjustments resulted in improved performance. Detailed implementation specifics are available in the the GitHub repository.

As shown in Table 10, Llama3.1-8B achieved the highest F1 score (98.8%), followed closely by Mistral-7B-v0.1 (98.2%). T5-base ranked third (97.5%), outperforming GPT-3.5 Turbo (94.7%). The superiority of Llama3.1-8B and Mistral-7B-v0.1 is even more evident in exact match scores. Llama3.1-8B correctly matched 1,819 out of 2,000 instances (90.9%), while Mistral-7B-v0.1 achieved 1,756 exact matches (87.8%), clearly surpassing T5-base (84.6%) and GPT-3.5 Turbo (68.2%).

These results are consistent with previously observed error patterns. Both T5 and GPT-3.5, which had earlier exhibited challenges with entity identification, now appear to effectively utilize the provided entities. While T5 still trails Mistral-7B-v0.1 in exact matches, it remains competitive in overall accuracy. Llama3.1-8B outperforms all models, demonstrating higher F1 and exact match performance, while Mistral-7B-v0.1 also shows strong results, ranking second in both metrics. Furthermore, the findings indicate that lightweight encoder-decoder

[26]https://huggingface.co/datasets/awalesushil/DBLP-QuAD?row=89

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Table 10. Performance metrics of the best models on DBLP-QuAD dataset: i) T5-base fine-tuned, ii) Mistral-7B-v0.1 fine-tuned, iii) GPT-3.5 Turbo employing Similarity with 7 shots, and iv) Llama3.1-8B fine tuned.

	T5-base	Mistral-7B-v0.1	GPT-3.5 Turbo	Llama3.1-8B
Precision	0.9774	0.9873	0.9567	0.9907
Recall	0.9793	0.9820	0.9519	0.9897
F1 Score	**0.9746**	**0.9820**	**0.9473**	**0.9885**
Bleu 4	0.9479	0.9607	0.8837	0.9744
Bleu C	0.9530	0.9646	0.8996	0.9774
Rouge 1	0.9884	0.9894	0.9733	0.9920
Rouge 2	0.9786	0.97887	0.9410	0.9846
Exact matches	1,693	1,756	1,364	1,819

models such as T5, which can be fine-tuned for translating natural language into SPARQL, exhibit strong potential as scalable, resource-efficient, and effective solutions for this task. However, the outstanding performance of Llama3.1-8B and Mistral-7B-v0.1 suggests that large-scale autoregressive models, when properly optimized, can be equally or even more effective, particularly when integrated with an entity resolution mechanism.

6.2 Execution-Based Evaluation on DBLP-QuAD

The metrics used so far to assess the model's performance have relied on comparing the query generated by the model with the expected query from the gold standard. An alternative evaluation approach involves executing the queries on a SPARQL endpoint and verifying whether the returned results match the expected output. This method has the advantage of allowing alternative formulations of the query that may differ syntactically but yield the same results, and are therefore equally valid from a practical perspective.

This type of evaluation was not feasible for SciQA, as most of its queries are not supported by data on a corresponding SPARQL endpoint. In contrast, DBLP-QuAD is backed by a well-maintained SPARQL endpoint, which enables the use of Execution-Based Evaluation. Specifically, its test set covers 2,000 queries, the vast majority of which (1,847) yield non-empty results when executed against the corresponding SPARQL endpoint.

In this section, we present and discuss the execution-based evaluation metrics for DBLP-QuAD. To this end, we executed the test queries from the DBLP-QuAD dataset against the official SPARQL endpoint[27] (QLever Server) and assessed the equivalence of the results produced by the model-generated and gold-standard queries. Specifically, we verified whether both query types yielded the same boolean value (in the case of `ASK` queries) or identical bindings (for `SELECT` queries). We also recorded instances in which the generated queries returned empty results or triggered execution errors, such as syntax issues or the use of unsupported SPARQL constructs in QLever. This analysis enabled us to identify different types of execution-time errors.

Table 11 presents the evaluation results. The *Exact Matches* row reports the number of generated queries that are identical to the gold queries at the string level. The *Non-Exact Matches & Correct Results & ≠ Empty* row includes cases in which the generated query differs syntactically from the gold query but returns the same non-empty result set. The *Non-Exact Matches & Correct Results & Empty* row refers to cases where both the generated and gold queries return the same empty result set, despite syntactic differences. The *Correct Queries* row aggregates all instances in which the generated query yields the same result as the gold query, regardless of syntactic similarity. Finally, *Execution Accuracy* [75] is a standard execution-based metric that measures the proportion of *Correct Queries* out of the total number of queries

The last two rows correspond to two distinct types of errors. The *Wrong Results* row indicates the number of queries that executed successfully but returned incorrect results. The *Execution Errors* row represents queries that

[27] https://sparql.dblp.org/

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failed to execute because of malformed syntax or the use of unsupported SPARQL constructs. By construction, the sum of the last three rows equals the total size of the test set, which covers 2,000 queries.

Comparing the number of *Exact Matches* with the *Correct Queries* - i.e., all queries that return the correct answer regardless of syntactic similarity - we observe a consistent trend. Notably, the two best-performing models in terms of exact matches, Llama3.1-8B and Mistral-7B-v0.1, are also the only models in this group capable of generating queries that differ syntactically from the gold standard yet still yield correct results. Specifically, we identified 75 such queries generated by Mistral-7B-v0.1 and 43 by Llama3.1-8B. It is also worth noting that Mistral-7B-v0.1 and Llama3.1-8B produced 114 and 69 queries, respectively, which were executable but returned incorrect results. This indicates that, in these cases, the models misinterpreted the original questions but were still able to construct syntactically valid, albeit semantically inaccurate, queries.

Overall, LLaMA 3.1-8B achieved the highest number of *Correct Queries*, with 1,871 instances (93.5% Execution accuracy), followed by Mistral-7B with 1,839 (91.9%). T5-base demonstrated moderate performance, yielding 1,693 matches (84.6%), whereas GPT-3.5 Turbo performed significantly worse, with only 1,364 matches (68.2%). These findings indicate that the previous evaluation, which relied solely on exact match, underestimated the practical performance of the models, as the best models can generate a substantial number of queries that retrieve correct answers even when they do not exactly match the gold standard.

Table 11. Execution-based evaluation on DBLP-QuAD.

	T5-base	*Mistral-7B-v0.1*	*GPT-3.5 Turbo*	*Llama 3.1-8B*
Exact Matches	1,693	1,756	1,364	1,819
Non-Exact Matches & Correct Results & ≠ Empty	0	75	0	43
Non Exact-Matches & Correct Results & Empty	0	8	0	9
Correct Queries	1,693	1,839	1,364	1,871
Wrong Results	0	114	0	69
Execution Errors	307	47	636	60
Execution Accuracy	84.65%	91.95%	68.20%	93.55%

6.3 Hybrid Retrieval-Augmented Majority-Based Answer Selection

Considering the results of the queries is not only valuable for evaluation purposes, but can also support the development of sophisticated retrieval-augmented approaches. For example, it is possible to build models that compare the result sets produced by multiple models, with the aim of achieving higher overall performance than any individual classifier.

To investigate this direction, here we propose and evaluate an ensemble method that employs a retrieval-augmented answer selection strategy. This approach selects the best answer from the top four models by applying a combination of majority voting and a heuristic based on output-type preference ranking. Specifically, given the outputs from the top four language models (T5-base, Mistral-7B, GPT-3.5-Turbo, and LLaMA 3.1-8B) on DBLP-QuAD, we implemented a per-query majority voting mechanism that selects the most frequent output among them. Since these outputs may differ not only syntactically but also structurally, for instance, returning booleans from `ASK` queries, lists from `SELECT` queries, or error messages, we normalised each candidate result using JSON serialisation to enable robust comparison. In 108 out of 2,000 queries, no single output received a majority of votes. In such cases, we ranked the candidate results based on their return type, following this priority: 1) non-empty results, 2) empty results, and 3) SPARQL error messages. If multiple candidates remained tied after applying both frequency and return-type preference, the final selection was made according to a fixed

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model priority: LLaMA, Mistral, T5, and GPT-3.5. This ordering reflects their previously measured performance and serves as a deterministic tie-breaking strategy.

Table 12 reports the evaluation of the retrieval-augmented approach using both exact match and execution-based metrics. The reported metrics are consistent with those shown in Table 11. Notably, the Retrieval-Augmented strategy achieved the highest overall performance, attaining a *Correct Queries* score of 1,898 (Execution Accuracy: 94.90%). This surpasses the performance of all individual models, as reported in Table 11. It also resulted in the fewest execution errors (41) and result discrepancies (61), thereby confirming its robustness.

A more in-depth analysis of Retrieval-Augmented ensemble solutions is beyond the scope of this paper. However, we consider this effort a promising first step that could be further explored by the research community in future work.

Table 12. Execution-based evaluation of the retrieval-augmented hybrid approach combining outputs from the top four models (T5-base, Mistral-7B, GPT-3.5-Turbo, and LLaMA 3.1-8B).

	Retrieval-Augmented Method
Exact Matches	1,836
Non-Exact Matches & Correct Results & ≠ Empty	53
Non Exact-Matches & Correct Results & Empty	9
Correct Queries	1,898
Wrong Results	61
Execution Errors	41
Execution Accuracy	94.90%

6.4 Transfer learning

We aimed to also investigate whether training a model on a specific benchmark for scientific question answering via natural language to SPARQL translation would enhance its performance on a different benchmark. To test this hypothesis, we trained models on one of the two benchmarks adopted in this study, SciQA and DBLP-QuAD, and evaluated their performance on the other. This research question is particularly relevant to the field of transfer learning [29, 76], which refers to a model's ability to leverage knowledge acquired from one task to improve its performance on a related task.

In this context, transfer learning refers to a model's ability to learn how to translate scientific questions into SPARQL queries within a dataset that follows a specific question style and then apply this capability to a different dataset with a distinct question formulation. However, an additional challenge arises: while a model may become proficient at composing syntactically correct SPARQL queries, it does not necessarily acquire knowledge of the specific ontological model required to retrieve accurate answers for a given dataset. For example, SciQA requires familiarity with the ontology used in the ORKG knowledge graph. To ensure a fair comparison, it is essential to distinguish between a model's ability to generate well-formed SPARQL queries and its ability to utilize specific relations and entity names to represent concepts accurately. DBLP-QuAD addresses this issue by explicitly providing the entities and relationships to be used, whereas SciQA does not. To mitigate this limitation and enable a fair comparison in a transfer learning setting, we created a modified version of SciQA, labelled SciQA_er in the following, that also provides the correct entities and predicates as input. We achieved this by analyzing the SPARQL queries of the SciQA dataset and extracting the entities and predicates from expressions in the WHERE clause whose prefix starts with *orkg*.

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Furthermore, we investigated whether training a model on a combination of the training sets from both benchmarks would lead to improved performance on either test set. If successful, this would suggest that assembling a diverse training set, even if derived from different domains and using different schemas, could contribute to better model performance.

To explore both direct transfer learning and training on a combined dataset, we conducted an additional set of experiments in which we varied the training and test sets for Llama3.1-8B and T5. These models were selected because they were, respectively, the best-performing models among encoder-only and encoder-decoder architectures.

Tables 13 and 14 present the performance results for different training and test set combinations. The experiments were labelled and conducted as follows:

- **sciqa-sciqa**: The model is fine-tuned on the SciQA dataset and evaluated on the same dataset.
- **dblp-dblp**: The model is fine-tuned on the DBLP-QuAD dataset and evaluated on the same dataset.
- **dblp-sciqa**: The model is fine-tuned on the DBLP-QuAD dataset and evaluated on the SciQA dataset.
- **dblp-sciqa_er**: The model is fine-tuned on the DBLP-QuAD dataset and evaluated on SciQA_er.
- **both-sciqa_er**: The model is fine-tuned on the training set of both DBLP-QuAD and SciQA datasets and evaluated on SciQA_er.
- **both-dblp**: The model is fine-tuned on both DBLP-QuAD and SciQA datasets and evaluated on the DBLP dataset.
- **sciqa_er-dblp**: The model is fine-tuned on SciQA_er and evaluated on the DBLP-QuAD dataset.

Table 13. Transfer learning. Performance metrics of Llama3.1-8B across various training and test set configurations.

	sciqa-sciqa	dblp-dblp	dblp-sciqa	dblp-sciqa_er	both-sciqa_er	both-dblp	sciqa_er-dblp
Precision	0.9793	0.9907	0.1929	0.6583	0.9861	0.9910	0.3670
Recall	0.9765	0.9897	0.0870	0.3430	0.9829	0.9872	0.5733
F1_Score	0.9765	0.9885	0.1070	0.4410	0.9835	0.9872	0.4261
Blue_4	0.9613	0.9744	0.0006	0.0111	0.9679	0.9714	0.1485
Bleu_Score_Cumulative	0.9637	0.9774	0.0013	0.0465	0.9712	0.9744	0.2211
Rouge_1	0.9796	0.9920	0.2144	0.5908	0.9887	0.9927	0.4783
Rouge_2	0.9681	0.9846	0.0437	0.2628	0.9780	0.9848	0.2569
exact matches	492	1,819	0	0	492	1,815	0
exact matches %	95.91%	90.95%	0.00%	0.00%	95.91%	90.75%	0.00%
total # of sample	513	2,000	513	513	513	2,000	2,000

It should be noted that training on the vanilla version of SciQA and testing on DBLP-QuAD yields mostly nonsensical results, as the latter requires the ability to process input that includes not only textual data but also entities and relations. Training on DBLP-QuAD and testing on the original SciQA resulted in poor performance, with an F1 score below 11% for both models. Training on DBLP-QuAD and testing on SciQA_er yielded a clearly better result for Llama3.1-8B (44.1% F1) but not for T5 (9.0% F1). This may suggest that Llama3.1-8B, possibly due to its greater ability to generalize, can better leverage transfer learning in this context. Training on SciQA_er and testing on DBLP-QuAD led to similar findings, with Llama3.1-8B (42.6% F1) again outperforming T5 (35.7% F1). These findings suggest that some degree of transfer learning is feasible in this domain. However, the results remain significantly lower than those obtained when training on a dataset that follows the same question style, SPARQL query structure, and schema of the test set.

The experiments that combined the training sets also produced interesting results. Training on the combination of both datasets and testing on SciQA_er resulted in a slight improvement in F1 scores for both Llama3.1-8B

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Table 14. Transfer learning. Performance metrics of T5-base across various training and test set configurations.

	sciqa-sciqa	dblp-dblp	dblp-sciqa	dblp-sciqa-er	both-sciqa-er	both-dblp	sciqa_er-dblp
Precision	0.9767	0.9774	0.1538	0.1435	0.9814	0.9779	0.3293
Recall	0.9760	0.9793	0.0600	0.0768	0.9781	0.9749	0.4561
F1_Score	0.9751	0.9746	0.0826	0.0902	0.9782	0.9730	0.3572
Blue_4	0.9597	0.9479	0.0031	0.0006	0.9626	0.9429	0.0950
Bleu_Score_Cumulative	0.9631	0.9530	0.0052	0.0018	0.9655	0.9491	0.1650
Rouge_1	0.9790	0.9884	0.1454	0.1972	0.9830	0.9865	0.5646
Rouge_2	0.9683	0.9786	0.0419	0.0339	0.9720	0.9750	0.3236
exact matches	483	1,693	0	0	485	1,652	0
exact matches %	94.15%	84.65%	0.00%	0.00%	94.54%	82.60%	0.00%
total # of sample	513	2,000	513	513	513	2,000	2,000

and T5 compared to the results obtained on the original datasets. Training on both datasets and testing on DBLP-QuAD yielded results comparable to those obtained when training exclusively on the original DBLP-QuAD dataset, with a slight decrease in F1 scores in both cases. These results indicate that combining datasets can yield benefits but may also introduce noise. Consequently, further research is required to identify optimal strategies for effectively integrating diverse datasets in this domain.

6.5 Resources and Computational Costs

We conducted fine-tuning experiments on Google Colab using the following virtual machine configurations. For both the T5-base and Dolly-v2-3B models, the VM was equipped with a T4 GPU, 2 vCPUs, and 12 GB of RAM. For the Mistral-7B-v0.1 model, the VM utilized an L4 GPU, while the CPU and RAM specifications remained consistent with the other models. For Llama3.1-8B, the VM was equipped with a T4 GPU, 2 vCPUs, and 12 GB of RAM. The fine-tuning process for T5-base took approximately 3 hours to complete 20 epochs on the DBLP-QUAD dataset and under 1 hour on the SciQA dataset. For Dolly-v2-3B, fine-tuning on the T4 GPU required about 5 hours on the SciQA dataset. The Mistral-7B-v0.1 model, running on the L4 GPU, took 14 hours to complete 10 epochs on the DBLP-QUAD dataset and less than 6 hours on the SciQA dataset. Both the Dolly-v2-3B and Mistral-7B-v0.1 models were quantized to 4 bits. The fine-tuning process for Llama3.1-8B took approximately 17 hours to complete 10 epochs on the DBLP-QUAD dataset and 4 hours on the SciQA dataset. The model was quantized to 4 bits.

7 LIMITATIONS

While this study provides valuable insights into the performance of LLMs on the SciQA benchmark, several limitations should be considered when interpreting the results and drawing conclusions.

Firstly, the focus of this work is specifically on the SciQA benchmark, which emphasizes scholarly artifacts and presents a unique challenge for QA systems. Therefore, the findings and conclusions may not be directly applicable to other QA benchmarks in the field. For instance, benchmarks that are completely manually curated, such as those mentioned in previous studies [42], are likely to pose different challenges since they are likely to contain more repeated patterns than SciQA. While SciQA provides a diverse set of questions and queries, it is important to note that the benchmark itself was primarily generated automatically from seed questions. This automated creation process may introduce biases or patterns that could potentially be exploited by LLMs, affecting the generalizability of the findings to real-world scenarios with more diverse and nuanced queries.

Another potential limitation is dataset contamination. Given that ORKG SciQA was only published in May 2023 and considering the pre-training times required for large models, there is a relatively low risk that the

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benchmark itself was seen during pre-training of LLM. However, we cannot rule out that similar queries to those in the SciQA benchmark were used or published before the official Nature dataset publication of ORKG SciQA.

A further limitation is that the benchmark differs from other QA benchmarks in that the performance of approaches depends strongly on the correct usage of literals rather than entities in the queries. For this reason, we do not employ an entity linker in our approach. Note that we could integrate an entity linker by feeding both the entity linker output and the natural language question to the LLM as done in e.g. [42] but this does not result in significant changes for SciQA and hence was omitted.

In addition to the aforementioned limitations, further research could explore additional factors that may impact the performance of LLMs on QA tasks, such as the robustness of LLMs to adversarial examples in the context of QA, and the scalability of the proposed methodologies to larger knowledge graphs or (even) more complex question types. Addressing these limitations and exploring these avenues for future research would contribute to a more comprehensive understanding of the capabilities and limitations of LLMs for question answering via semantic parsing.

8 CONCLUSIONS

This paper conducted a thorough analysis of how LLMs perform on the SciQA benchmark. We assessed six distinct language models using three methodologies: zero-shot learning, few-shot learning, and fine-tuning. Furthermore, we investigated eight different strategies for selecting examples in few-shot learning. Our results indicate that optimizing LLMs through appropriate prompting and fine-tuning techniques leads to outstanding performance on this benchmark. The best model achieved an accuracy of over 95.9% in exact query matches, which is usually an underestimate of actual performance, and an F1-score exceeding 97.6%. These results underscore the necessity for more challenging benchmarks in this domain.

The fine-tuned Mistral-7B-v0.1 achieved the highest F1 scores, while the fine-tuned Llama3.1-8B attained the highest number of exact matches. Despite its significantly smaller size of 220 million parameters, T5 also delivered highly competitive performance. These models were followed by GPT-3.5 utilizing a seven-sample few-shot approach, by the fine-tuned Dolly-v2-3b, and the fine-tuned GPT-2. This suggests that smaller, low-resource models, when carefully fine-tuned, can compete with the performance of larger, resource-intensive models, even in the context of a complex task like scientific question answering. The analysis also showed that semantic similarity [47] seems to be the most efficient method for sample selection in few-shot learning for this task. The error analysis revealed that Llama3.1-8B, T5, and GPT-3.5 struggled the most with comprehending the underlying ontological schema, whereas Mistral-7B-v0.1 exhibited a higher rate of semantic misunderstandings, which are errors arising from an incorrect interpretation of the input questions.

We then performed further experiments on the DBLP-QuAD benchmark, which provides relevant entities and relations, and demonstrated that Llama3.1-8B and T5 perform exceptionally well when supplied with this type of information. We also evaluated the top-performing models using execution-based metrics, which indicate that their practical performance is better than what is reflected by the exact match metric. Subsequently, we investigated a retrieval-augmented ensemble approach and demonstrated that it outperforms all individual models on the *DBLP-QuAD* dataset. We believe this approach is highly promising and warrants further investigation in future work.

Finally, we conducted experiments to assess the potential of transfer learning in this domain. Specifically, we examined whether training a model on a specific benchmark or a combination of them could improve its performance on a given test set. The results suggest that a certain degree of transfer learning is feasible in this context. However, further research is required to identify the most effective strategies for integrating heterogeneous datasets and optimizing knowledge transfer.

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An important consideration is that current LLMs achieve very high performance on the present generation of benchmarks, both when considering exact match and execution-based metrics. While this demonstrates the excellent progress of research in this field and the rapid improvement of LLMs, it also underscores the urgent need for more challenging benchmarks to further advance methods in this domain.

For our upcoming research endeavors, we aim to extend our investigation by conducting thorough experiments to assess the proficiency of LLMs in knowledge graph question answering across diverse domains. We also plan to explore ensemble approaches [20], LLM collaboration frameworks [48], code auto-generation solutions [73], and the pre-training of small models from scratch. We also aim to investigate the potential of integrating LLMs with knowledge graphs for related tasks, such as formulating scientific hypotheses [11] and generating literature reviews [9]. Finally, we intend to leverage the valuable insights acquired from this study to craft a more demanding benchmark for scientific question answering.

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